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Professor Gianfranco Chicco
Editor-in-Chief, *Sustainable Energy, Grids and Networks* (SEGAN)
Elsevier

Subject: Submission of Original Research Article to *Sustainable Energy, Grids and Networks*

Dear Professor Chicco and Editorial Board,

I am pleased to submit my original research manuscript titled:

"PULSE: Adaptive Event-Triggered Inertia and Frequency Support from Data Center UPS Fleets"

for consideration for publication in *Sustainable Energy, Grids and Networks* (SEGAN).

Context and Motivation

In low-carbon power grids, the progressive displacement of synchronous thermal generation by inverter-based renewables severely degrades natural system inertia. Data centers house multi-megawatt electrochemical uninterruptible power supply (UPS) batteries that remain idle for over 99% of their operating life. This manuscript introduces **PULSE** (**P**rimary-frequency **U**PS **L**ocal **S**ub-second **E**vent-triggered control), a communication-aware decentralized framework that mobilizes these distributed battery fleets to provide synthetic inertia and fast frequency support without requiring centralized dispatch.

Alignment with SEGAN Aims and Scope

This work directly addresses core topics within the journal's scope:

- **Control of Distributed Energy Storage**: Decentralized coordination of distributed rack-level battery assets under strict local State-of-Charge (SoC) reserve constraints.
- **ICT Integration and Cyber-Physical Systems**: Event-triggered communication over shared local networks subject to packet loss, variable latencies, and jitter.
- **Grid Dynamics and Stability**: Rigorous frequency-domain delay margin formulations coupled with power-frequency swing dynamics.
- **Sustainability (UN SDGs 7 and 11)**: Unlocking pre-capitalized computing infrastructure for clean grid stabilization, displacing carbon-intensive fossil peaker reserves.

Core Contributions and Key Findings

- **Zero-Communication Idle State & 63.6% Traffic Reduction**: PULSE employs an adaptive, time-relaxing event threshold that sends zero standby messages during normal ambient frequency drift, while cutting contingency communication traffic by 63.6% compared to fixed-threshold schemes.
- **Realistic Measurement Filtering**: Evaluated under physical 1-cycle PLL ($65.0\,\mathrm{ms}$) and 2-cycle PMU ($90.5\,\mathrm{ms}$) low-pass filters under measurement noise, achieving smooth, non-saturating derivative-droop injection without inverter chattering.
- **Frequency Stabilization & Inertia Support**: In a 300 MVA benchmark, 200 distributed agents ($12\,\mathrm{MW}$) arrest a 5% generation trip at $59.721\,\mathrm{Hz}$ (vs. $59.423\,\mathrm{Hz}$ unsupported), providing a $+41.7\%$ virtual inertia boost and improving the 500 ms sliding-window RoCoF from $-0.441\,\mathrm{Hz/s}$ to $-0.285\,\mathrm{Hz/s}$.
- **Provable Delay Robustness**: Linear frequency-domain stability analysis proves an analytical delay margin of $1.58\,\mathrm{s}$ ($0.952\,\mathrm{s}$ under nominal filtering), verified against severe packet drop and latency jitter.
- **Guaranteed Data Center IT Reliability**: Delivering a full 30 s frequency response consumes only $0.166\,\mathrm{kWh}$ per rack, maintaining over 40 minutes of autonomous backup runtime and ensuring 100% compliance with Uptime Institute Tier III/IV standards.

Declarations

- **Sole Authorship & Originality**: The author is the sole contributor. The manuscript is original and has not been submitted or published elsewhere.
- **Conflict of Interest**: The author declares no competing financial interests or personal relationships.
- **Data Availability**: All simulation code, verification suites, and reproduction scripts are publicly available on GitHub at `https://github.com/mahmoudsajjadi/PULSE` and permanently archived on Zenodo (DOI: 10.5281/zenodo.14920188).
- **AI Statement**: AI-assisted tools were used solely for language editing and grammar polishing. The author reviewed and independently verified all contents.

Thank you for your consideration of this manuscript.

Sincerely,

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